Problem

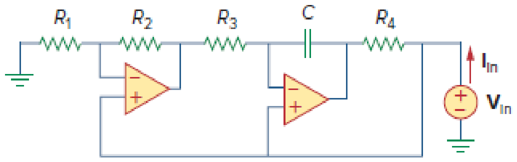
The op amp circuit in Fig. 10.131 is called an *inductance simulator*. Show that the input impedance is given by

$$\mathbf{Z}_{in} = \frac{\mathbf{V}_{in}}{\mathbf{I}_{in}} = j\omega L_{eq}$$

where

$$L_{eq} = \frac{R_1 R_3 R_4}{R_2 C}$$

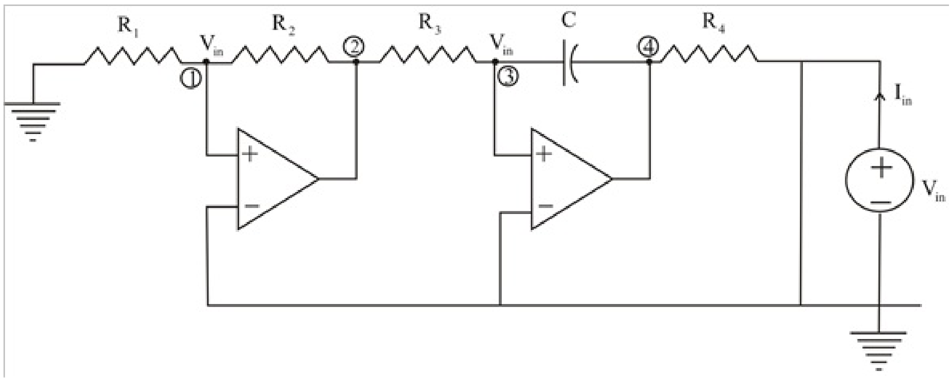
Figure 10.131:



Step-by-step solution

Step 1 of 4

Consider the circuit below.



Step 2 of 4

At node 1,

$$\frac{0 - V_{in}}{R_1} = \frac{V_{in} - V_2}{R_2}$$

$$\frac{-V_{in}}{R_1} = \frac{V_{in} - V_2}{R_2}$$

$$-V_{in}(R_2) = (V_{in} - V_2)R_1$$

$$-V_{in} + V_2 = \left(\frac{R_2}{R_1}\right)V_{in} \quad \text{..................} \quad (1)$$

Step 3 of 4

At node 3,

$$\frac{V_2 - V_{in}}{R_3} = \frac{V_{in} - V_4}{\frac{1}{j\omega C}}$$

$$\frac{V_2 - V_{in}}{R_3} = j\omega C\left(V_{in} - V_4\right)$$

$$-V_{in} + V_4 = \frac{V_{in} - V_2}{j\omega C\, R_3} \text{ .... (2)}$$

Step 4 of 4

From (1) & (2),

$$-V_{in} + V_4 = \left(\frac{-R_2}{j\omega C\, R_3\, R_1}\right) V_{in}$$

Thus,

$$I_{in} = \frac{V_{in} - V_4}{R_4}$$

$$I_{in} = \left(\frac{R_2}{j\omega C\, R_3\, R_1\, R_4}\right) V_{in}$$

$$Z_{in} = \frac{V_{in}}{I_{in}} = \frac{j\omega C\, R_1\, R_3\, R_4}{R_2}$$

Therefore,

$$Z_{in} = j\,\omega\, L_{eq}$$

Where,

$$L_{eq} = \frac{R_1\, R_3\, R_4\, C}{R_2}$$